

Bocconi

COMPUTER SCIENCE

Code 30424 a.y. 2024-2025

Lesson 3

Introduction to the spreadsheet



Università
Bocconi
MILANO

Objectives of the lesson

- Understand the basic principles for managing a dataset
- Understand how a spreadsheet works
- Learn the correct approach to using Excel
- Review the knowledge required to attend the course

Definition of dataset

A dataset is a collection of data, gathered and organized in a matrix format (**Tables**), in which each column represents a different variable (**Fields**) and each row corresponds to a specific element of the collection (**Record**)

Creating a dataset

Starting from the collected data, with different characteristics and different formats, in order to build a dataset on which to carry out the processing it is necessary to proceed with their consolidation

To this end, it is necessary to carry out on the available data a process of **data cleaning, integration** and **homogenization**

Spreadsheet

It is the most widely used tool today, both in individual and corporate settings, for consolidating, processing, and analyzing data.

Main features:

- Accessibility (intuitive and easy-to-use interface)
- Interactivity (automatic recalculation)
- Versatility (hundreds of mathematical, statistical, logical functions, etc.)
- Completeness (analysis tools: pivot tables, scenarios, solver, etc.)

What is a spreadsheet

A spreadsheet is basically a double entry grid consisting of:

- Cells
- Rows
- Columns
- Worksheets
- Workbooks

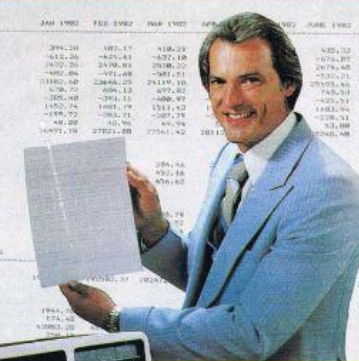
C11 (L) TOTAL

	A	B	C	D
	ITEM	NO.	UNIT	COST
1	MUCK	43	12.95	556.85
2	BUZZ	15	6.75	101.25
3	TOE	250	49.95	12487.50
4	EYE	2	4.95	9.90
			SUBTOTAL	13155.50
			9.75% TAX	1282.66
			TOTAL	14438.16

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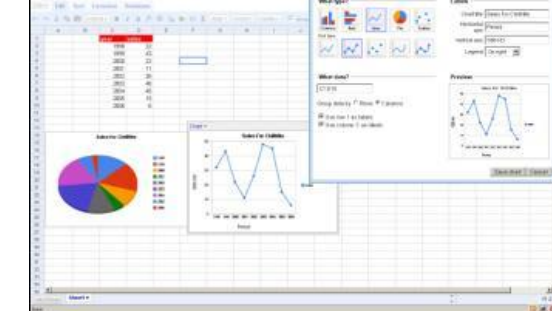
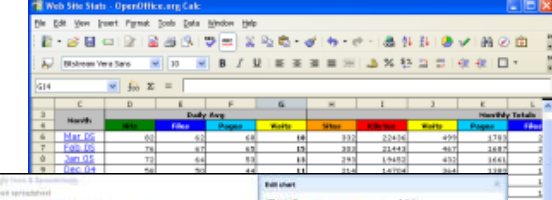
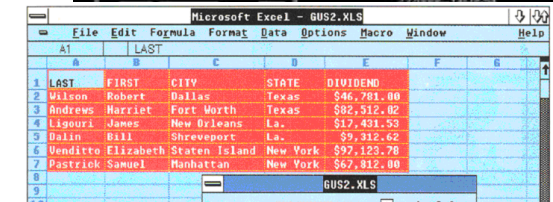
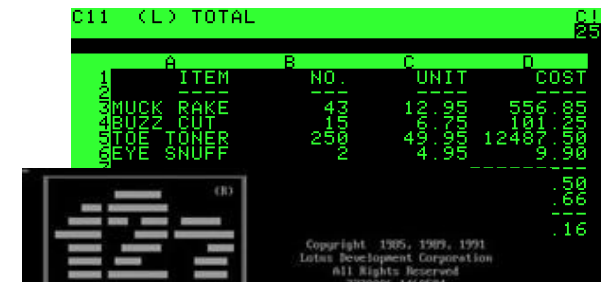
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The evolution of the spreadsheet

- The concept of spreadsheet was introduced in 1961 by Richard Mattessich
- In 1978 VisiCalc was released, designed by Dan Bricklin
- In 1983 Lotus 1-2-3 was born
- In 1985 Microsoft Excel was born
- In 2000 OpenOffice.org was born with the Calc spreadsheet
- In 2006 Google presented Web Docs & Spreadsheets usable on any browser



Excel basics

The basic concepts that are important to have clear in order to work easily with Excel concern the following features:

- A **workbook** is a set of overlapping **worksheets**
- Each worksheet is divided into **rows** and **columns**
- Rows are indicated by numbers
- Columns are indicated by letters
- The intersections of rows and columns form the **cells**
- Each cell is indicated by a letter (column to which it belongs) and a number (row to which it belongs) that make up its **reference**
- The name of the **worksheet** and of the **file** can also be part of a cell reference
- Each cell can contain **data** or **instructions** (i.e. formulas and functions)

Data in Excel

The categories of data that can be managed with Excel are:

- Text strings
- Numerical data
- Logical values



This includes the data types seen in lesson 2.
How they are represented depends on the regional settings of the Operating System



Correct use of a spreadsheet

To use Excel correctly and professionally, the following guidelines should be taken into account:

1. Formatting must always be functional to the clear organization of data (they should not be considered as "aesthetic solutions")
2. It is good practice to leave a blank line (at the bottom of the table) before inserting any summary elaboration (e.g. totals, averages, ...) so as to avoid that these values appear as additional records of the dataset
3. It is important to maintain the consistency of the dataset in terms of: organization and arrangement of data, formatting, processing, etc.
4. The creation of formulas and functions must take into account the principles of efficiency and optimization. For example, using a syntax that makes the formula/function applicable to all cells where the same type of processing is required
5. It is essential to use the "\$" symbol correctly in cell references

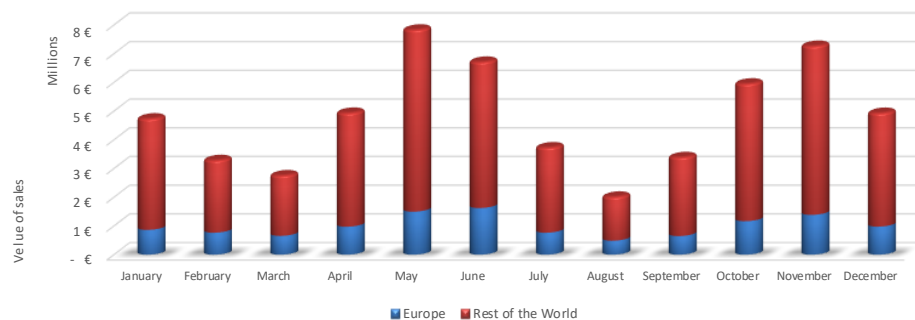
Exercise

Using the knowledge acquired with the ICDL prerequisite, open the file "30424 lesson 3.xlsx" and complete it with the formatting and processing necessary to obtain the result shown in the «Sample» worksheet.

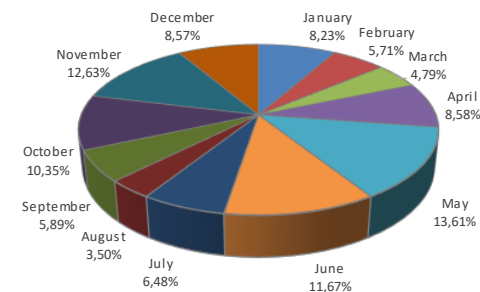
Sales last year

	Europe				Rest of the World				Worldwide			
	Amounts	% on the territory yearly sales	% on worldwide monthly sales	% on worldwide yearly sales	Amounts	% on the territory yearly sales	% on worldwide monthly sales	% on worldwide yearly sales	Amounts	% on the territory yearly sales	% on worldwide monthly sales	% on worldwide yearly sales
January	864.254 €	7,32%	18,25%	1,50%	3.871.858 €	8,47%	81,75%	6,73%	4.736.112 €	8,23%	100,00%	8,23%
February	764.523 €	6,47%	23,26%	1,33%	2.522.926 €	5,52%	76,74%	4,38%	3.287.449 €	5,71%	100,00%	5,71%
March	654.789 €	5,54%	23,75%	1,14%	2.101.873 €	4,60%	76,25%	3,65%	2.756.662 €	4,79%	100,00%	4,79%
April	973.468 €	8,24%	19,72%	1,69%	3.962.015 €	8,66%	80,28%	6,88%	4.935.483 €	8,58%	100,00%	8,58%
May	1.497.638 €	12,68%	19,12%	2,60%	6.335.009 €	13,85%	80,88%	11,01%	7.832.647 €	13,61%	100,00%	13,61%
June	1.625.462 €	13,76%	24,21%	2,82%	5.087.696 €	11,12%	75,79%	8,84%	6.713.158 €	11,67%	100,00%	11,67%
July	764.319 €	6,47%	20,49%	1,33%	2.965.558 €	6,48%	79,51%	5,15%	3.729.877 €	6,48%	100,00%	6,48%
August	486.427 €	4,12%	24,15%	0,85%	1.527.381 €	3,34%	75,85%	2,65%	2.013.808 €	3,50%	100,00%	3,50%
September	648.376 €	5,49%	19,12%	1,13%	2.742.630 €	6,00%	80,88%	4,77%	3.391.006 €	5,89%	100,00%	5,89%
October	1.167.538 €	9,89%	19,61%	2,03%	4.786.906 €	10,47%	80,39%	8,32%	5.954.444 €	10,35%	100,00%	10,35%
November	1.386.738 €	11,74%	19,08%	2,41%	5.879.769 €	12,86%	80,92%	10,22%	7.266.507 €	12,63%	100,00%	12,63%
December	976.482 €	8,27%	19,80%	1,70%	3.954.752 €	8,65%	80,20%	6,87%	4.931.234 €	8,57%	100,00%	8,57%
Total	11.810.014 €	100,00%	20,52%	20,52%	45.738.372 €	100,00%	79,48%	79,48%	57.548.386 €	100,00%	100,00%	100,00%

Sales last year



Worldwide sales last year



Total sales last year:	57.548.386 €
Montly average Europe	984.168 €
Lowest value of monthly sales	486.427 €

Tools to learn with today's lesson

- Editing the contents of a worksheet (managing rows, columns, ...)
- Formatting (borders, filling, data format, ...)
- Formulas (linking cells, using the "+" and "/" operators)
- Functions (SUM, AVERAGE, MIN, MAX, IF)
- Inserting and editing a chart
- Managing worksheets (unhide, hide, insert, delete)



Book references

Excel for students in economics and finance, 2nd edition:

Chapters 1.1, 1.2.1, 1.2.2, 1.2.3, 1.2.4, 1.3.1, 1.4.4, 1.5.1, 1.5.2, 1.5.3, 2.2

Excel workbook, 4th edition:

Exercises on the functions and tools of the lesson